

Name: _____ Partner: _____ Period: _____

How Many Answers Does This Question Have? An equation is a question, and the honest answer is sometimes “every number” and sometimes “none.” With your partner, decide *what kind of question* each item is before you touch it — then answer it, and check. Argue about the ones you disagree on before writing anything down.

Tier R — Remediate

Part 1 — Solve, and name one move. Solve each equation. In the last column, name *one* property of equality you used.

Equation	Solution	One property you used
(a) $3x + 7 = 22$	$x =$ _____	_____
(b) $2(x - 4) = 10$	$x =$ _____	_____
(c) $5x - 3 = 2x + 12$	$x =$ _____	_____
(d) $\frac{x}{4} + 2 = 7$	$x =$ _____	_____

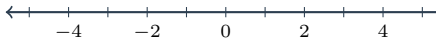
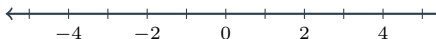
Pick any one and check it by substituting into the original. Which one, and does it check?

Part 2 — Read the last line only. Four students finished four different equations. You are given only their *final* line. Say how many solutions each equation has. Do not solve anything.

Final line	How many solutions?	How you knew
(a) $x = 3$	_____	_____
(b) $5 = 5$	_____	_____
(c) $0 = 7$	_____	_____
(d) $-2 = -2$	_____	_____

In (b), (c), and (d) the variable disappeared. Is that a mistake? _____ What decides which answer you give? _____

Part 3 — Solve and shade. Solve, then shade the solution set. Watch your circle: open $\circ$ for $<$ and $>$, closed $\bullet$ for $\leq$ and $\geq$.

(a) $2x + 1 < 7$	x _____	
(b) $-4x \geq 8$	x _____	

One of those two required you to reverse the symbol. Which, and why?

Part 4 — Quadratics that factor. Every one of these factors cleanly. Set one side to **zero** first if it is not already.

- (a) $x^2 - 9 = 0$, $x =$ _____ (b) $x^2 + 6x + 8 = 0$, $x =$ _____
- (c) $x^2 - 7x + 12 = 0$, $x =$ _____ (d) $x^2 + 5x = 0$, $x =$ _____

Check yourself. Substitute *one* of your answers to (b) into $x^2 + 6x + 8$. Do you get zero?

Every quadratic here had *two* answers. Item (a) is the one people get half right — why?

Tier A — Approaching Proficiency

Part 1 — Mixed, and nobody is telling you which kind it is. Decide first, then work.

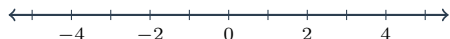
1. $\frac{2x}{3} - 1 = \frac{x}{2} + 2$

(Multiply every term by 6 first.)

2. Solve for x : $ax + 3x = b$.

The variable appears *twice*. What did you have to do before you could divide?

3. $8 - 2x \leq 14$



4. $x^2 = 4x + 21$

5. $x^2 + 6x + 4 = 0$ — nothing factors. Use the formula and **simplify the radical**.

Is your answer rational or irrational? _____

Part 2 — Two situations, answered in sentences.

6. A parking garage charges \$4 to enter plus \$2.50 per hour. You have \$20.

Inequality: _____ Solve: h _____

Answer the question that was asked, in a sentence: _____

7. A ball is thrown upward. Its height in feet after t seconds is $-16t^2 + 64t$. When is the ball 48 feet high?

Equation: _____ $t =$ _____

You got *two* answers, and both are real times. Explain what is physically happening at each one. _____

Part 3 — Predict, then solve. For $2x^2 - 4x - 3 = 0$:

$b^2 - 4ac =$ _____ Prediction: _____ real solutions Now solve: $x =$ _____

Did the count match your prediction? _____ Why is predicting first worth anything at all? _____

Tier E — Extension

Part 1 — Put the parameter inside the equation. Until now the number of solutions was something you discovered *after* solving. Make it a condition instead.

Number of solutions **For which k does $kx + 6 = 3x + 6$?** **For which k does $kx + 6 = 3x + 9$?**

exactly one _____

no solution _____

infinitely many _____

One box in each column has *no* possible k . Say which, and why that case simply cannot happen for that equation. _____

Part 2 — Spot the error. Each student made **exactly one** mistake. **Name** what went wrong — naming it is the part that counts — and give the correct answer.

Student's work	What went wrong (name it)	Correct
(a) $-2x > 10 \Rightarrow x > -5$	_____	_____
(b) $(x - 3)(x + 2) = 6$, so $x - 3 = 6$ or $x + 2 = 6$	_____	_____
(c) $x^2 = 25 \Rightarrow x = 5$	_____	_____
(d) $3(x - 4) = 3x - 12$: “the x ’s cancel, so no solution”	_____	_____

Two of these four are failures to *interpret* rather than failures to compute. Which two?

Part 3 — The discriminant as a condition. Consider $x^2 - 6x + c = 0$, where c can be any real number.

(a) Write the discriminant in terms of c : _____

(b) Two real solutions when c _____; one real solution when c _____; no real solutions when c _____

(c) You just answered a question about *equations* by solving an *inequality*. Say in one sentence why that was the right tool. _____

Part 4 — Why zero and nothing else. The zero product property says $mn = 0$ forces $m = 0$ or $n = 0$. There is no version of it for $mn = 8$. Explain why not — and use the word **factors**. _____
